

▣ 11 Applications of probability

Statistics and probability is applied to address many practical problems that have a stochastic aspect. We briefly look at three such applications here.

- Queuing theory
- Reliability
- SPC (statistical process control)

▣ Queuing theory

What is a queuing system?

Everybody has experience of queuing scenarios from queuing for a taxi/ATM/supermarket checkout etc. Queuing systems also exist in less obvious situations such as traffic attempting to pass through a signal-controlled junction or (computer) processes attempting to access the CPU on a multitasking computer system.

Models exist that allows us to determine certain, steady state characteristics of particular queuing systems.

The M/G/1 queue

Example

Customers arrive to use an ATM (automatic telling machine) at an average rate of 20 per hour and this arrival pattern can reasonably well be described with a Poisson distribution.

The time taken by customers to use the ATM has mean 2 minutes and standard deviation 36 seconds.

The M/G/1 model can be used to analyse this queuing system provided the following assumptions are met.

- Customers do not renege (decline to join) or baulk (leave after joining)
- The arrival and service timing parameters do not change
- Customers arrive independently (i.e., not in pairs or groups)
- The ATM does not break down
- The system operates indefinitely
- **N.B.** The average service time is strictly less than the average inter-arrival time.

A number of steady-state (i.e., long term) characteristics of the queuing system can be calculated using the following formula.

ρ	P_0	L_q	L_s	W_q	W_s
$\frac{\lambda}{\mu}$	$1 - \rho$	$\frac{(\lambda \cdot \sigma)^2 + \rho^2}{2 \cdot (1 - \rho)}$	$L_q + \frac{\lambda}{\mu}$	$\frac{L_q}{\lambda}$	$W_q + \frac{1}{\mu}$

Analyse this queuing system

Solution

Any convenient time unit can be used – here hours seems suitable

λ is the arrival rate and is 20 customers per hour.

μ is the service rate and is 30 customers per hour.

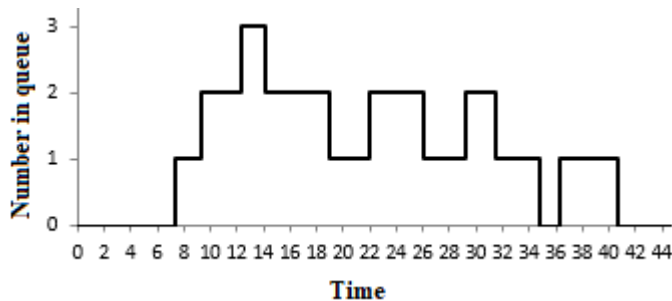
σ is the standard deviation for the service time is 36 seconds or $^{36}_{/3600} = 0.01$ hours

$\rho = 0.67$ is the server utilisation (proportion of time the ATM is busy)

$P_0 = 0.33$ proportion of time server is idle.

$L_q = 0.727$ customers is the average length of the queue

The queuing system changes at discrete points in time – at such times the number in the queue will increase or decrease by 1. Thus, a time series chart of the number in the queue might look like the following.



An average queue length of 0.727 means that this is the average height of this chart

$L_s = 1.39$ customers is the average number in the system (in the queue OR using the ATM)

$W_q = 0.036$ hours (2.18 minutes) is the average waiting time in the queue

$W_s = 0.07$ hours (4.18 minutes) is the average waiting time in the system

Example

A firm produces bespoke kitchens to order with each order taking an average of 11 days (SD 2.24 days). Orders are received randomly in time (i.e., according to an exponential distribution) with an average time between orders of 2 weeks. Show that the firm will be idle over 20% of the time, the average number of jobs in the queue will be about $1\frac{1}{2}$ and customers may expect to have to wait about 32 weeks from when they place an order until the order is complete.

Solution

Take weeks as the time unit

λ ?

Two weeks / order $\equiv \frac{1}{2}$ order / week

$\lambda = 0.5$

μ ?

11 days / order $\equiv \frac{1}{11}$ orders / day $\equiv \frac{7}{11}$ orders / week

$\mu = 0.636$

σ ?

2.24 days $\equiv \frac{2.24}{7} = 0.32$

$\rho = 0.786$ and $P_0 = 0.214$

So firm is idle 21% of the time

$L_q = 1.5$ customer

$W_s = 4.57$ weeks or 32 days

M/M/c queue

An M/M/c queue assumes that the service time as well as the inter-arrival times can be modelled with an exponential distribution and that there are c servers.

An exponential service time in effect implies that the length of time already spent on service has no effect on the length of the remaining service time.

This “memoryless” property rarely applies to task times – if we spend any time on a task, we will be closer to completing the task than we were at the start. For example, if it takes 30 minutes to mow a lawn and a gardener has already spent 10 minutes on this task, we wouldn’t expect the time to complete the task to remain at 30 minutes.

But there are some tasks where this memoryless property might be seemed to be applied. A couple of (contrived!) examples are given below.

Example

The time to the observation of the next squirrel by a zoologist walking through a forest.

Example

The time between a HR manager, who is perusing CVs submitted for a post, locating suitable candidates.

Sometimes this “memoryless” property might be viewed as approximately applied. For example, the time to diagnose a problem or fault – why won’t a machine start, or the cause of a bug in a software application.

Even if the task time can in no way be construed as memoryless and if the task time is highly variable, then an exponential task time might make for a reasonable model. For example, the time taken by a doctor to attend to a patient or the time taken by a helpline operator to answer a customer query.

✎ *Remember that all mathematical models are simplifications and therefore approximations of real-world systems or process. It is not sensible to ask if a model is a precise representation of the system (it never is!) – rather you should ask if the model is adequate for the purpose at hand.*

It is left as an exercise to find the formulae for calculating steady-state values. Websites can readily be found that will carry out the calculations – for example <https://www.supositorio.com/rcalc/rcalclite.htm>

Example

A technical support call centre has 40 operators. The time taken to answer a call can be modelled as exponential with mean time of 12 minutes and a customer call is made on average every 20 seconds.

Show that each operator will be idle for an average of 48 minutes in an 8-hour day, the average number of waiting calls will be almost 40 and that customer must wait for a little over one minute before they can speak with an operator.

On one particular day, three operators are absent and there are only 37 working. The impact on the quality of customer service will be to increase average queue length to over 65 and average queuing time to about 10 min.

Exercise

On another day, the full complement of 40 operators are working but the demand increases so that a call arrives every 18.2 seconds rather than every 20 seconds. Show that the average waiting time before a customer speak to an operator increases from about 1.25 minutes to over 25 minutes.

□ Reliability

Definition

The reliability of a component or system is the probability that it will be able to perform its function when required, for a specified time hours (the design life) and in a particular environment.

Example

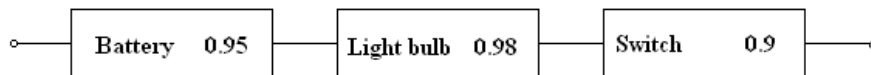
A hiker takes a torch on a trip. The torch consists of three components whose lives are suitably modelled to give reliability measures, taking the total usage time of the torch during the trip duration to be 6. The components are the battery, the light bulb and the switch which have reliability measures of 95%, 98% and 90% respectively.

What is the reliability of the flash lamp?

We assume the events of individual components failing are statistically independent.

Solution

Clearly the system can only work if all the components work (i.e., there is no redundancy). Thus, we arrange our components in series.



To determine the system reliability, we take the product of the component reliabilities. This holds for any number of components. I.e., $R_{\text{system}} = 0.95 \cdot 0.98 \cdot 0.90 = 0.8379$ or about 84%

This rule follows immediately from the assumption that the components fail independently and from the laws of probability.

N.B. The overall reliability of components arranged in series is always less than the reliability of any one of the individual components.

How might the measures of reliability been obtained?

The normal density makes for a reasonable probability model to describe the survival time of the battery. Suppose the mean and standard deviation are found to be 6.8 and 0.486 hours respectively.

Then the reliability of the battery = $\Pr(T > 6)$ where T is the survival time of the battery and $T \sim N(6.8, 0.486^2)$

$$= \text{pnorm}(6, 6.8, 0.486, \text{lower.tail}=\text{FALSE}) \sim 0.95$$

The failure of the bulb might be adequately described as a Poisson process with a mean time between failure (MTBF) of 300 hours.

Thus, the reliability of the bulb = $\Pr(F = 0)$, where $F = \#$ failures in 6 hours and $F \sim \text{Poi}(6/300)$

$$= \text{dpois}(0, 6/300) \sim 0.98$$

The switch might be used an average of 125 times every 6 hours and the manufacture of this switch might have established using a large scale test that 10% of such components will fail after being switched 125 times.

Thus, 90% might be expected to be operational after 125 used – hence this is the reliability.

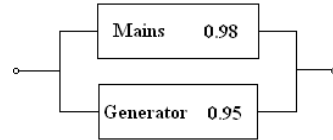
Example

A hospital has a main electricity supply and an independent backup generator. There is a 2% chance that in one year, the mains supply will fail and a 5% chance that the generator will fail. How reliable is the hospital's electricity supply?

Solution

Our components mains supply (M) and generator (G) are such that for a design life of one year, $R(M) = 0.98$ and $R(G) = 0.95$. Clearly there is a degree of redundancy here as, unlike our flashlight example, we can permit one of the components to fail, provided that not both fail.

We can arrange our system using the RBD shown opposite. This, for obvious reasons, is called a parallel arrangement.



The reliability here is $R_{\text{sys}} = 1 - (1 - R_{\text{Mains}})(1 - R_{\text{Generator}}) = 0.999$

The rule again follows immediately from the assumption that the components fail independently and from the laws of probability.

I.e., $R_{\text{sys}} = \text{Pr}(\text{system does not fail})$

$= 1 - \text{Pr}(\text{system does fail})$

$= 1 - \text{Pr}(\text{generator fails AND Mains fail}) = 1 - \text{Pr}(\text{generator fails}) \text{Pr}(\text{Mains fail})$ *Independence*

$= 1 - (1 - \text{Pr}(\text{generator does not fail})) (1 - \text{Pr}(\text{mains does not fail})) = 1 - (1 - R_{\text{Generator}})(1 - R_{\text{Mains}})$

Note

This holds for any number of components. For example, for 3 components in parallel with reliabilities R_1 , R_2 and R_3 the reliability of the system is $1 - (1 - R_1)(1 - R_2)(1 - R_3)$ etc.

N.B. The overall reliability of components arranged in parallel is always more than the reliability of any one of the individual components.

Example

A utility company electronic data on both its customers and on matters relating to its own internal affairs. All of this information is retained on a secure server.

A backup policy is implemented whereby each night, all data is burned on to portable drives, with two such devices used – one to store the customer data and one for company data. This process is repeated for two further drives. The four drives are taken home by four employees (who live in different areas) and the devices are reused the following night.

In addition, three cloud-based, backup services are used. All data is backed up to CloudSafe, the customer data only is backed up to BackupOnline and the company data is backed up to DataStore. Three passwords (changed daily) are used – one for all the portable drives, one for the cloud-based services and one for the secure server.

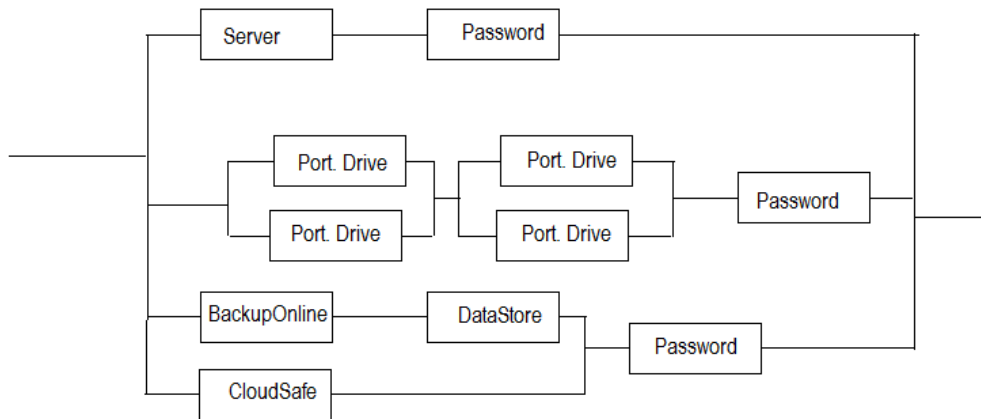
Suitable probability models are applied to estimate the probability of a drive / service not failing overnight – these reliability estimates are tabulated below. In addition, the probability of a password being forgotten can be taken to be 0.1%

Component	Server	Portable drive	CloudSafe	BackupOnline	DataStore
Reliability	0.99	0.98	0.9999	0.999	0.998

The IT manager is asked to quantify the risk of the company losing (any) data and suggests that such an event would happen about once every N days, what is N?

Solution

To find N we must draw the RBD and then use the rules for serial and parallel arrangements to simplify the diagram until we get a single component.



Exercise

How would the value of N change if

- (i) The same password was used to access all devices / services?
- (ii) Different passwords were used to access all devices / services?

N.B. there is an order of magnitude of difference between a reliability measure of 0.999 and 0.9999. The former amounts to a failure one time in 1,000, the latter one time in 10,000,

□ Statistical process control (SPC)

Most products are manufactured to customer specified dimensions or properties. However, random variation (caused by differences in raw materials, differences in operator techniques etc.) will lead to some deviation from these target values.

For example, a food producer may have a specification that the concentration of sugar in a food product should be 3g per 100g, but the actual sugar content would typically vary around this target value, even if there is nothing amiss with the production run – i.e., even if the process is in control.

A difficulty that arises with most such processes is that they sometimes go out of control (OOC).

For example, if a machine used in the process is incorrectly configured or accidentally adjusted etc., the actual mean sugar content in the product might change to 3.3.

A quality controller (QC) might be employed to monitor the process by taking samples at regular intervals and measuring the sugar concentration.

It is unlikely that the measured concentration will be exactly 3. The QC might obtain a value of 2.96 or 3.14 and she then needs to choose between two possible explanations as to why the measured sample concentration differs from the target value of 3.

Is it because of random variation inherent in such processes? If so, the process is still in control and no intervention is needed.

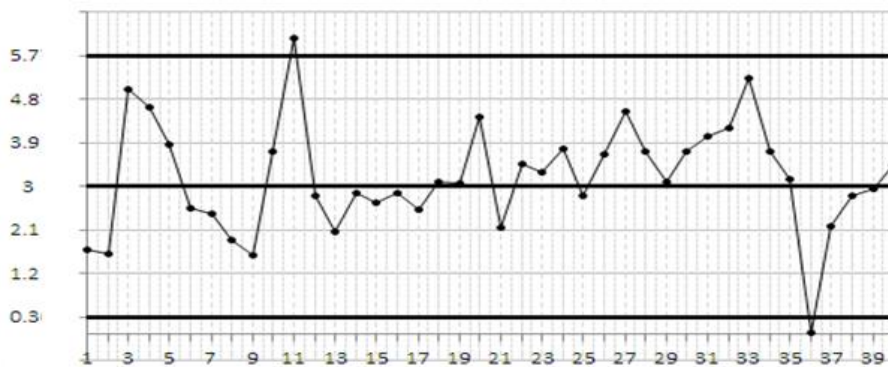
Or is it because the process mean has actually changed? If so, the process has gone out of control and frequently this will necessitate that the process be stopped, and the cause of the problem identified.

As with any binary decision, there are two possible mistakes that might be made. The process might be deemed to be out of control when it is actually in control. Or it may be deemed to be in control when in reality it is out of control.

In general, it is not possible to eliminate the risk of making one or other of these errors but in some cases, it is possible to measure the likelihood of the error.

Control charts

A control chart is a time series chart used to monitor a process. In our example here, the QC may have decided to take a sample from the process every (say) 15 minutes, calculate the sugar concentration for that sample and plot this value (as a dot) on a chart. For ease of reading, these dots are connected with line segments. The sample control chart below monitors the process for 10 hours.



The line at the centre of the chart is the central line (CL) and corresponds to the target mean of 3. The other two solid lines are the upper and lower control limits. These are set at a distance of 3 standard deviations from the CL (the standard deviation need to be known/estimated for the process – here is it 0.9)

We will assume that the quantity of interest (sugar concentration) follows a normal distribution as is usually the case with most processes.

The QC would analyse this signal to decide if the process is out of control. The following four patterns are often interpreted as indicating that the process is out of control.

- | | |
|---|-----------------|
| 1. A single value outside the control limits. | E.g., sample 11 |
| 2. 9 points in a row on one side of the central line | E.g., sample 34 |
| 3. 6 points in a row, all increasing or all decreasing. | E.g., sample 9 |
| 4. 14 points in a row, alternating up and down. | E.g., sample 26 |

A useful calculation might be to ask how likely is it that such a signal would have been generated had the process stayed in control. I.e., how likely is a false OOC signal?

Signal 1?

Using R, we find the probability of a reading within 3 SD of the mean is

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pnorm(5.7, 3, 0.9) - pnorm(0.3, 3, 0.9) = 0.99732
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and so, the probability of being more than 3 SD from the mean is about 0.0027 (0.27%)

Signal 2?

If the process were in control, then a sample value is as likely to be above the mean as below. Thus, 9 on one side of the mean would have the same likelihood of flipping a coin, finding that it is heads – then considering how likely is it that in the next 8 flips we also get heads.

The probability of this is easily obtained to be $0.5^8 = 0.0039$

Signal 3?

We say the rank of the sequence {4.7, 2.1, 9.2, 3.6} is 3, 1, 4, 2 because 4.7 is the 3rd smallest number, 2.1 is the 1st smallest number, 9.2 is the 4th smallest number and 3.6 is the 2nd smallest number.

A similar calculation for a sample of 6 successive concentration values would yield some permutation of the integers 1- 6, with the values 1, 2, 3, 4, 5, 6 corresponding to a “run up” and 6, 5, 4, 3, 2, 1 a “run-down”

As the number of permutations of the integers 1 – 6 is given by $6! = 720$ and if we assume that all of these permutations are equally likely (*Exercise: Are they?*) then the likelihood of one of the two OOC signals arising by chance is given by $2/720 = 0.00277$

Signal 4?

Exercise

Hint: If you cannot estimate it analytically, try writing a simulation in R.